Lesson 32: Area of Polar Region

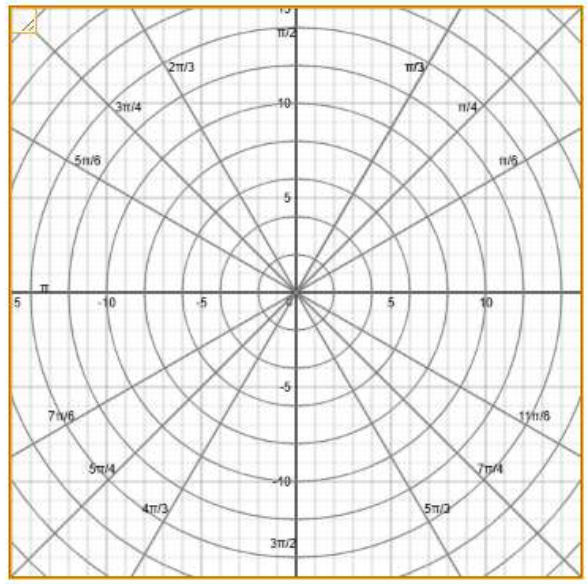
1) Find the area of the interior of $r = 2\sin\theta$.

a) Graph $r = 2\sin\theta \qquad 0 \le \theta \le \pi$

b) When $\theta = \pi/2$, $r =$ ___?___

c) When $\theta = \pi$, $r =$ ___?___

d) Find Area $= \dfrac{1}{2}\displaystyle\int_{\alpha}^{\beta}[f(\theta)]^2 d\theta = \dfrac{1}{2}\int_{\alpha}^{\beta} r^2 d\theta =$ ___?___



2) Find the area of one petal of $r = 4\sin 2\theta$.

a) Graph $r = 4\sin 2\theta \qquad 0 \le \theta \le 2\pi$

b) Note: $\sin 2\theta = r/4 \quad \Rightarrow \quad 2\theta = \sin^{-1}(r/4)$

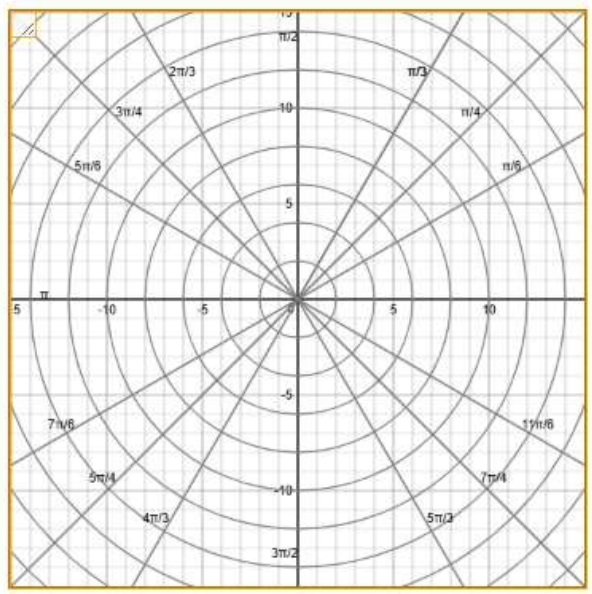
When $r = 0$,

$2\theta = \sin^{-1}(0) \qquad \theta =$ ___?___

$2\theta = \pi - \sin^{-1}(0) \qquad \theta =$ ___?___

$2\theta = \sin^{-1}(0) + 2\pi \qquad \theta =$ ___?___

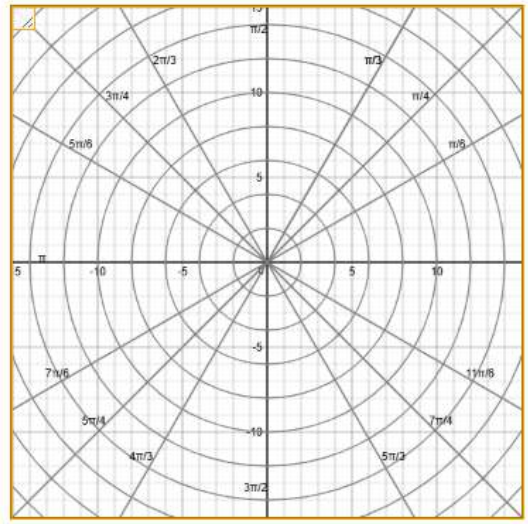
$2\theta = 3\pi - \sin^{-1}(0) \qquad \theta =$ ___?___



c) The petal in Quadrant I runs from $\theta =$ ___?___ to $\theta =$ ___?___.

d) Find Area $= \dfrac{1}{2}\displaystyle\int_{\alpha}^{\beta}[f(\theta)]^2 d\theta = \dfrac{1}{2}\int_{\alpha}^{\beta} r^2 d\theta =$ ___?___

3) Find the area of the interior of $r = 2 - \sin\theta$.



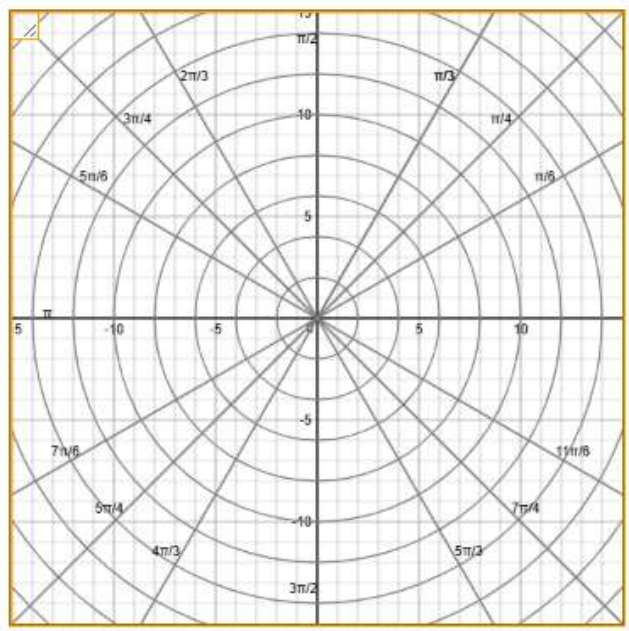
Hint: Graph of $r = 2 - \sin\theta$ does not touch the pole (0,0). Therefore, we cannot set $r = 2 - \sin\theta$ equal to zero. Instead use the Graphing Calculator to figure out α and β.

a) Graph $r = 2 - \sin\theta \qquad 0 \le \theta \le 2\pi$

b) Find the area of the interior of $r = 2 - \sin\theta$.

$$\text{Area} = \frac{1}{2}\int_{\alpha}^{\beta}[f(\theta)]^2 d\theta = \frac{1}{2}\int_{\alpha}^{\beta} r^2 d\theta = \underline{\quad ? \quad}$$

4) Find the area of one petal of $r = 4\cos 3\theta$.



a) Graph $r = 4\cos 3\theta \qquad 0 \le \theta \le \pi$

b) Note: $\cos 3\theta = r/4 \quad \Rightarrow \quad 3\theta = \cos^{-1}(r/4)$

When $r = 0$,

$3\theta = \cos^{-1}(0) \qquad \theta = \underline{\quad ? \quad}$

$3\theta = 2\pi - \cos^{-1}(0) \qquad \theta = \underline{\quad ? \quad}$

$3\theta = \cos^{-1}(0) + 2\pi \qquad \theta = \underline{\quad ? \quad}$

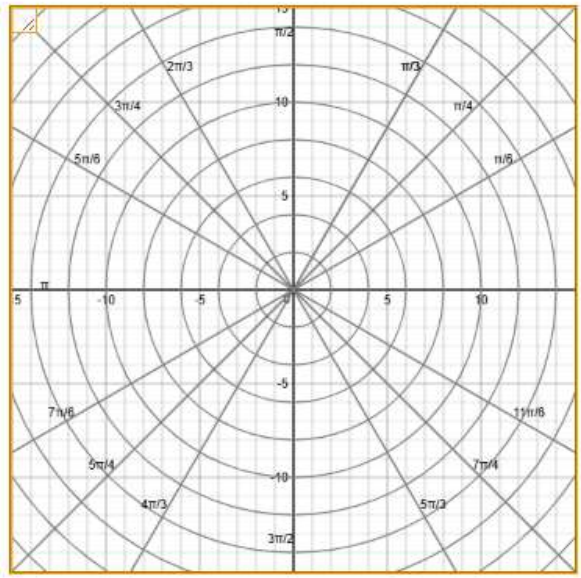
$3\theta = 4\pi - \cos^{-1}(0) \qquad \theta = \underline{\quad ? \quad}$

c) The petal in Quadrant I, runs from $\theta = \underline{\quad ? \quad}$ to $\theta = \underline{\quad ? \quad}$.

d) Area of one petal $= \frac{1}{2}\int_{\alpha}^{\beta}[f(\theta)]^2 d\theta = \frac{1}{2}\int_{\alpha}^{\beta} r^2 d\theta = \underline{\quad ? \quad}$

5) Find the area of the inner loop of $r = 1 + 2\sin\theta$

a) Graph $r = 1 + 2\sin\theta \qquad 0 \le \theta \le 2\pi$



b) Note: $\sin\theta = (r-1)/2 \quad \Rightarrow \quad \theta = \sin^{-1}[(r-1)/2]$

When $r = 0$,

$\theta = \sin^{-1}(-1/2) \qquad \theta =$ ___?___

$\theta = \pi - \sin^{-1}(-1/2) \qquad \theta =$ ___?___

$\theta = \sin^{-1}(-1/2) + 2\pi \qquad \theta =$ ___?___

$\theta = 3\pi - \sin^{-1}(-1/2) \qquad \theta =$ ___?___

c) Inner loop runs from $\theta =$ ___?___ to $\theta =$ ___?___.

d) Area of the inner loop $= \dfrac{1}{2}\displaystyle\int_{\alpha}^{\beta} [f(\theta)]^2 \, d\theta = \dfrac{1}{2}\displaystyle\int_{\alpha}^{\beta} r^2 \, d\theta =$ ___?___